

# IT314 - Lab 1

## Assignment: Git & GitHub Fundamentals using a Bug Fix Workflow

### Objective

The objective of this assignment is to gain hands-on experience with Git and GitHub by following a version control workflow. The emphasis of this lab is on correctly using Git and GitHub features.

### Problem Statement

You have been provided with a C++ solution for the problem **"Valid Parentheses."** The solution intentionally contains a bug. Your task is to identify the issue, fix it, and manage the complete development process using Git and GitHub.

### Assignment Requirements

1. Create a local Git repository containing the provided solution file.
2. Create a new GitHub repository and connect it to your local repository.
3. Push the initial version of the project to GitHub.
4. Identify the bug in the provided solution.
5. Create a GitHub Issue describing the identified bug.
6. Create a separate feature/fix branch to work on the issue.
7. Implement the fix in the solution file.
8. Commit your changes using a meaningful commit message.
9. Push the branch to GitHub.
10. Open a Pull Request from the feature branch to the main branch.
11. Merge the Pull Request after verifying the changes.
12. Ensure the final repository reflects a clean and meaningful Git history.

### Files Provided

- valid\_parentheses.cpp (contains an intentional bug)

### Submission

- GitHub repository link
- Screenshot of the GitHub Issues page
- Screenshot of the Pull Request page
- Screenshot of the final commit history
- Screenshot of the corrected program execution/output

### Important Notes

The Git workflow has already been demonstrated during the lab session through the demo video. This assignment evaluates your ability to apply that workflow independently.